

NETRA ALLE

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PROFESSIONAL SUMMARY

Final-year B.Tech student in Artificial Intelligence & Data Science at AISSMS IOIT, Pune; who approaches every problem depth-first — doesn't move on until the fundamentals are understood, the edge cases are handled, and the system actually works end-to-end. Proficient in Python, MLOps, NLP, and full-stack ML engineering. Driven by curiosity and a habit of going beyond surface-level solutions — currently deepening expertise in LLMs and RAG systems with the same thoroughness applied to every project. Building and deploying end-to-end AI systems — from data pipelines and model training to FastAPI serving, Docker containerisation, and CI/CD.

EDUCATION

AISSMS Institute of Information Technology, Pune 2023 – 2027 (Expected)

B.Tech, Artificial Intelligence & Data Science | CGPA: 8.09 | Full-Time

HSC (12th): 76.33% | Vidya Bhavan School and Jr. College | Maharashtra HSC Board | 2023

SSC (10th): 90.20% | St. Joseph's Convent High School | Maharashtra SSC Board | 2021

Relevant Coursework: Machine Learning, Deep Learning, NLP, Computer Vision, LLMs & Generative AI, Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Discrete Mathematics, Statistics, Artificial Intelligence.

TECHNICAL SKILLS

Languages: Python, SQL, C++ (Basics)

AI / ML: Machine Learning, Deep Learning, NLP, DistilBERT, HuggingFace Transformers, Scikit-learn, TF-IDF, Cosine Similarity, EDA, Time-Series Forecasting

LLM & GenAI: RAG Systems, HuggingFace Inference API, Prompt Engineering (learning)

MLOps / DevOps: MLflow, DVC, Evidently AI, FastAPI, Docker, GitHub Actions CI/CD, Prometheus, Streamlit

Cloud & Deploy: Render, Vercel, HuggingFace Hub, AWS Cloud Foundations

Frontend: React 19, Recharts (data visualisation)

Data & Viz: Pandas, NumPy, Matplotlib, Seaborn, Power BI, Tableau | Databases: MySQL, MongoDB

Core CS: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, DBMS, Problem Solving

Tools: Jupyter, VS Code, Git, GitHub, Cursor

PROJECTS

Fake News Detection – MLOps Pipeline | [GitHub](#) | [Live Demo](#)

Python | DistilBERT | HuggingFace | FastAPI | React 19 | Docker | MLflow | DVC | Evidently AI | Prometheus | GitHub Actions

- Fine-tuned DistilBERT on 42,826 balanced articles; identified and fixed a critical Reuters dateline bias — model was learning source signatures, not content signals. Documented honest gap between 99.9% training accuracy and 71% external validation rather than reporting only the flattering number.
- Built a complete MLOps pipeline: data ingestion → feature engineering → model training → FastAPI serving → Docker containerisation → React 19 frontend; deployed end-to-end on Render + Vercel with model hosted on HuggingFace Hub.
- Integrated Prometheus instrumentation (/metrics endpoint) for live API observability — request count, latency histograms, and custom domain counters (prediction class distribution, low-confidence rate, A/B model split).
- CI/CD via GitHub Actions with 17 real behavioral API tests (TestClient + unittest.mock) — tests JSON schema, value ranges, and edge cases without the 255MB model in CI; 97% code coverage on app.py.

Sales Analytics Pipeline | [GitHub](#) | [Live Demo](#)

Python | SQL | Pandas | Power BI | Time-Series Forecasting

- Engineered an end-to-end pipeline processing 9,800+ sales records — preprocessing, feature engineering, and structured storage in a relational SQL database.
- Built interactive Power BI and HTML dashboards surfacing product-level KPIs and regional revenue trends; implemented time-series forecasting validated with 20 automated unit tests — zero regressions.

CERTIFICATIONS

- AWS Forage – Solutions Architecture Virtual Internship (Amazon) | Jul 2025
- Microsoft – Fundamentals of Machine Learning | Mar 2025
- Deloitte – Cyber Security Job Simulation (Forage) | Jun 2025
- AWS Academy – Cloud Foundations | Feb 2026

ACHIEVEMENTS & ACTIVITIES

- **Research Publication** – "Artificial Intelligence for Detecting Algorithmic Manipulation in Social Media" — IJSREM Journal, Volume 10, Issue 6, Jun 2026.
- **InnovateYou Techathon 3.0 (National Level, 2026)** – Participated in 24-hour national-level hackathon at AISSMS IOIT Pune; Organised by Innovation Foundation, Pune (Feb 2026).
- **Volunteer, Jal Dindi Prathisthan NGO, Pune** – Led 5 environmental awareness drives (plastic cleansing, water conservation, nature trails); certified by Trustee Rajeev Bhawsar (Feb–Apr 2025).